

PAPER_1680 — Origin of Inertia Scale = $U_{inertia} = SO_5 = 10$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Foundational **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1466 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

$$U_{inertia} = SO_5 = 10$$

Status: **EXACT**.

UQFF Closed Identity

$$U_{inertia} = SO_5 = 10 \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical foundational measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1466
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "inertia_origin_10"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.